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Hatching of nile tilapia (*Oreochromis niloticus*) egg in the hatching medium use salinity media and bromelain enzyme

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Abstract. This research aimed to observe the success of Nile Tilapia egg hatching in the hatching medium and the survival rate of larvae used salinity media and bromelain enzyme. This study was handled at laboratory of aquaculture, Fisheries and Marine Science Faculty, Brawijaya University on January-March 2020. This research used experimental method, completely randomized design with five treatments and three replications for every treatment. The salinity media was A (0 ppt), B (2.5 ppt), C (5 ppt), D (7.5 ppt), and E (10 ppt), and for every treatment was added 0.1223 μmol bromelain enzyme. The results showed that the highest percentage of hatching rate was obtained from 5 ppt with an average of 88.89%, while the lowest hatching rate was obtained in 0 ppt with an average of 67.78%. The results of the observation of abnormalities showed that abnormalities occurred in each treatment, except 0 ppt. The highest survival rate was obtained in 5 ppt with an average of 95.15%, while the lowest percentage of survival was obtained in 10 ppt with an average of 76.62%. The water quality was classified as good. The temperature of the hatching medium is obtained in the range 24^o C - 27^o C.

1. Introduction

Tilapia is a euryhaline fish, which has a low risk in the cultivation process, low operating costs, even artificial feeding is carried out 15 days before harvest [1]. The amount of tilapia production is increasing due to the increasing demand for tilapia. Increasing the production of good tilapia each year still requires further development so that the production target is achieved.

The main problem in the development of tilapia aquaculture in brackish water is seed. Tilapia seeds must be adapted before stocking. The adaptation process of tilapia seeds from fresh water to saline water takes a long time. According to Djunaedi *et al.* [2], the adaptation process of tilapia seeds was carried out in stages by adding seawater to the seed reservoir gradually until it reached 5 ppt every week. It is less time efficient if the required salinity is higher.

The method that could be done to shorten the adaptation time of the seeds is by hatching tilapia eggs in saline media. Salinity affects egg hatching and larvae survival. Salinity affects the level of osmotic process, water absorption and the hardening process of the *chorion* membrane in eggs, so it can be assumed that this phenomenon will also affect the utilization of egg yolk energy for embryo growth and *osmoregulation* [3].

The process of hatching fish eggs can be accelerated using the protease enzyme. The protease enzyme will dissolve the egg wall (outer shell), so that the embryo will easily come out [4]. One of the protease enzymes is bromelain enzyme which is found in the Bromiliceae family of plants from

fruit, stems and leaves. This enzyme is included in the extracellular protease enzyme which can hydrolyze protein into simpler compounds such as short-chain peptides and amino acids [5].

In the previous studies, addition the enzyme bromelain to the hatching of tilapia had an effect on the hatching rate of tilapia eggs. Tilapia eggs hatched in different salinity media with media containing 0.1223 μmol of bromelain enzyme are expected to increase the hatching rate of tilapia eggs.

2. Materials and methods

2.1 Experiment method and design

This study consisted of 5 treatments and was repeated 3 times. The salinity treatments in this study were (A) 0 ppt, (B) 2.5 ppt, (C) 5 ppt, (D) 7.5 ppt and (E) 10 ppt and the media contained the enzyme bromelain 0.1223 μmol . The study was controlled using incubation hatches made from 320 ml bulb bottles.

2.2 Research procedures

2.2.1 Hatching incubator design.

Hatchery of tilapia using an incubator made from 320 ml bulb bottle. The bottle cap is perforated in the middle to place the aerator hose. The aerator hose is put into the bottle and placed on the edge. Before connecting the hose to the aerator, an air regulator is installed to adjust the aeration strength. The aeration is intended to allow a rotating current, so that the eggs continue to move. In addition, aeration is also used as an oxygen supply.

2.2.2 Container preparation and hatching media. Preparation begins with cleaning the 320 ml bulb bottle and the incubator tools by washing them with detergent, then drying them for one day. Preparation of salinity media using 1 L of fresh water mixed with rough salt (NaCl) without iodine. Preparation of salinity media by dissolving salt in water with a ratio of 1 gram / liter to produce 1 ppt, then weighed and calculated according to treatment. After that, 0.1223 μmol of bromelain enzyme is added. Enzymes are added using the dilution formula according to Prabowo *et al.* [6], namely:

$$V_1 \times C_1 = V_2 \times C_2$$

Information:

V1: Solution Volume 1

V2: Solution Volume 2

C1: Solution concentration 1

C2: Solution concentration 2

2.2.3 Spawning.

The spawning was conducted in a concrete pool measuring 1x3 m, then filled with clean water to a height of 75 cm. After that, commercial pellets were fed with a protein content of 28%. Feeding is done 2 times a day ad libitum in the morning and evening. Spawning was carried out en masse with a male and female ratio of 1: 3.

2.2.4 Preparation of Egg Sample .

Tilapia egg samples used in the study came from spawning at the Fish Reproduction Division of the Fish Cultivation Laboratory, Faculty of Fisheries and Marine Sciences, Brawijaya University, Malang. Egg samples were obtained from the mother's mouth of tilapia after spawning. According to Prabowo *et al.* [6], harvesting tilapia eggs can be done by capturing the female parent and then taking the eggs that are still incubated in the mouth. The method used is known as the "ketek system". After that, the eggs were divided by 30 eggs per treatment and put into each hatching incubator.

2.2.5 Maintenance of larvae.

The hatched larvae are transferred to plastic cups for maintenance until the yolks run out. The plastic cups are aerated as a source of oxygen. At the time of maintenance, there is no feeding, because it has food reserves in the form of egg yolk.

2.3 Parameters

2.3.1 Major parameters

2.3.1.1. Hatching rate.

The main parameter is the success tilapia (*Oreochromis niloticus*) eggs to hatch. According to Prabowo *et al.* [6], hatching rate is the percentage of eggs hatched that can be known and calculated using the formula, namely:

$$\text{HR (\%)} = \frac{\text{hatched egg}}{\text{Total egg}} \times 100\%$$

2.3.1.2. Abnormality

Abnormality is a form of deformity or deviation of the body organs (deformity). According to Mukti *et al.* [7], observing the abnormalities of tilapia larvae was carried out by observing the state of the tilapia larvae body shape including the shape of the head, body and tail.

2.3.1.3. Survival rate.

Information of survival rate parameters is gained after the maintenance of tilapia larvae for seven days or when the yolk of the tilapia egg has been completely absorbed. This measurement was made by comparing the presentation of the number of fish that survived during the maintenance time with the number of hatched eggs. According to Mukti *et al.* [7], the survival rate formula is showed below:

$$\text{SR (\%)} = \frac{\text{Larvae (t)}}{\text{Larvae (0)}} \times 100\%$$

2.3.2. Supporting Parameters

2.3.2.1. Water quality.

Water quality observed during the study included temperature, pH, DO (dissolved oxygen) 2 times a day, namely in the morning at 04.00 WIB and at 14.00 WIB. This time is estimated to have different fluctuations.

2.4 Data analysis

The data obtained were analyzed statistically using analysis of variance or the F test (ANOVA) according to a completely randomized design (CRD). This is done to determine the effect of treatment (independent variables) on the parameters analyzed. When the F test value is significantly different or very significant, then proceed with the least significant difference test (LSD) 5% and LSD (1%) and continue with the orthogonal polynomial test.

3. Results and discussion

3.1 Hatching rate

The results of observations on media with different salinity on the hatching rate of tilapia eggs showed that the hatching rate value of tilapia (*Oreochromis niloticus*) eggs in this study ranged from 66.7%-93.33%. The hatching rate of tilapia eggs will increase with increasing salinity, but at a certain point the hatching rate value reaches a maximum point and the yield decreases. The hatching rate percentage of tilapia fish eggs is presented in Figure 1.

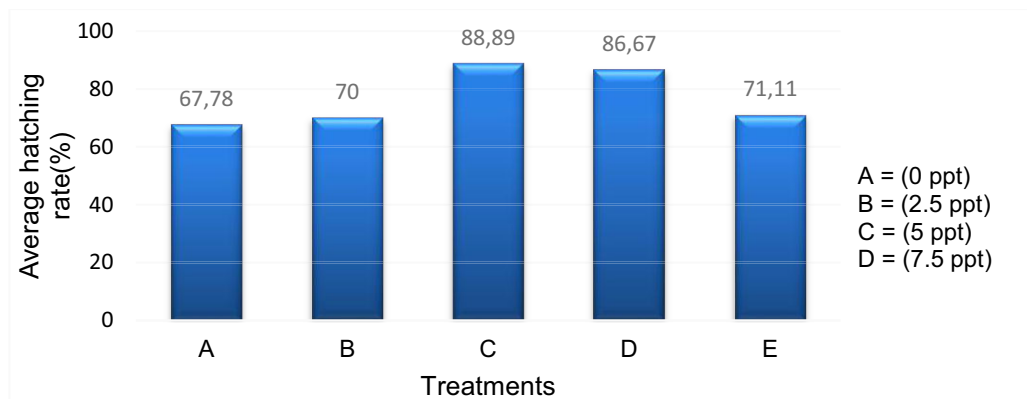


Figure 1. Average of hatching rate

The hatching rate graph shows that the treatment that produces the lowest hatching rate is treatment A (salinity 0 ppt), which is an average of $67.78\% \pm 1.92$, while the treatment that produces the highest hatching rate is treatment C (salinity 5 ppt) which is average of $88.89\% \pm 3.85$. The low hatching rate in treatment A was caused by fungal attack on the eggs. The fungus forms fine threads and covers the entire surface of the egg, so that the egg cannot develop and eventually die. This is in conformity with the explanation of Lingga *et al.* [8] declared that the spores of the fungus can penetrate the *chorion* layer in the egg, then develop and carry out the reproductive process by absorbing the nutrients in the egg. These spores will form *hyphae* which cause disruption of respiration of the eggs, so that the eggs will die. Eggs that are attacked by fungus can occur due to low temperatures. According to Rustadi [9], low water temperature is a suitable environment for the development and growth of fungi.

The high hatching rate in treatment C is due to the salinity of 5 ppt the egg conditions are close to the *isoosmotic* state, so that the energy required to maintain *osmose* pressure in the *osmoregulation* process can be used for embryo development. According to Fujaya [10], *isoosmotic* is a condition between the concentration of body fluids (eggs) and the concentration of the media. The farther the difference in *osmose* pressure between the body and its environment, the greater the metabolic energy required to carry out *osmoregulation*. In addition, at 5 ppt salinity there was no fungal attack on the eggs. These results are different from the results obtained by Darwisito *et al.* [11], which states that the optimal salinity for hatching tilapia eggs without bromelain enzyme is 10 ppt. At the salinity of 0 ppt and 2.5 ppt there are still fungal attacks. This is consistent with the statement of Jones and Jennings [12] that the growth of freshwater fungi in sea water media is lower than in freshwater media.

The results of the variance test showed that the hatching of fish eggs using saline media with the administration of bromelain enzymes showed a very significant effect with F value greater than 1% F value. The relationship (regression) between the salinity media treatment and the hatching rate value can be calculated using the orthogonal polynomial test. The regression graph of the effect of salinity on the hatching rate of tilapia eggs is presented in Figure 2.

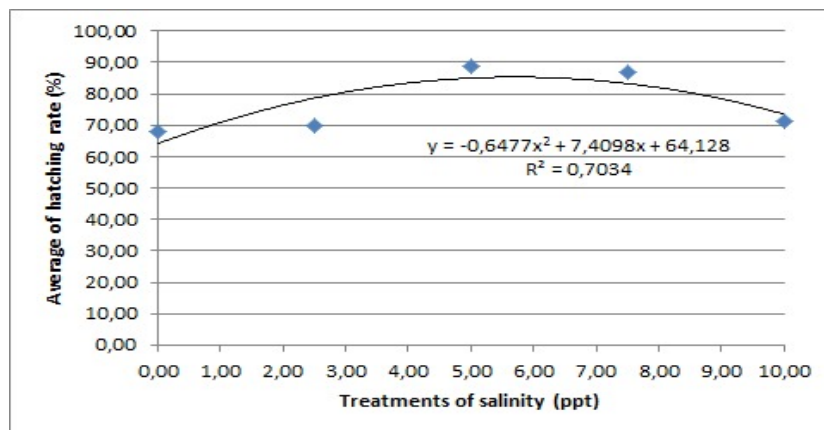


Figure 2. Regression between salinity media treatment with hatching rate

The graph results obtained in the form of a quadratic curve so that the higher the salinity, the higher the hatching rate obtained. However, at a certain point the hatching rate reaches a peak and the yield will decrease further.

3.2 Abnormality

Observation of abnormalities is needed to determine how much influence the difference in salinity media and the addition of 0.1223 μmol of bromelain enzyme to the percentage of defective tilapia larvae. The abnormalities observed showed that abnormalities occurred in almost every treatment, except A treatment (salinity 0 ppt).

Treatment E (10 ppt salinity) produced the highest larvae abnormality (3.04%), this was due to the salinity of 10 ppt tilapia fish eggs could not tolerate this salinity so that the development of tilapia body organs was inhibited. This is in accordance with the statements of [13], that the higher the salinity, the greater the potential for disruption of egg development. Exposure to high salinity and outside tolerance limits can cause tissue formation to be disrupted, so that organ refinement stops.

Abnormalities that occur in this study are generally in the form of spinal disorders. According to Luthfiyah *et al.* [14] stated that the environmental movement of tilapia from fresh water to brackish water can have a negative impact on the cells that make up tilapia vertebrates. When the salinity of the medium is higher (*hyperosmotic*) to the eggs, the water in the eggs will come out, so the eggs will shrink and when they hatch the eggs will be abnormal.

The difference in morphology of abnormal tilapia larvae with normal larvae is shown in Figure 9. The abnormality that occurs is Caudal Deformity Syndrome (CDS). According to Tave *et al.* [15], fish with CDS experienced an upward curvature of the spine from the tail region and did not develop fully. Fish with this deformity cannot swim efficiently. Comparison of photos of abnormal larvae and normal larvae can be seen in Figure 3.

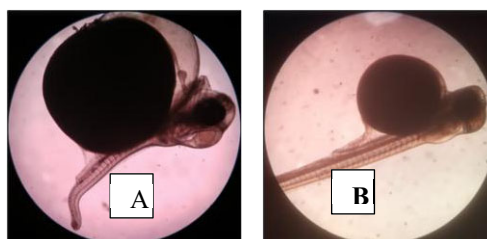


Figure 3. Abnormal fish larva (A), normal fish larva (B) (40x)

3.3 Survival rate

Survival rate observations are carried out when the yolk of the larvae has run out. The results of the survival observations of tilapia showed that the percentage level of survival of the larvae in this study was > 76.62% on average. The average of the percentage of survival rate of larvae can be noticed in Figure 4.

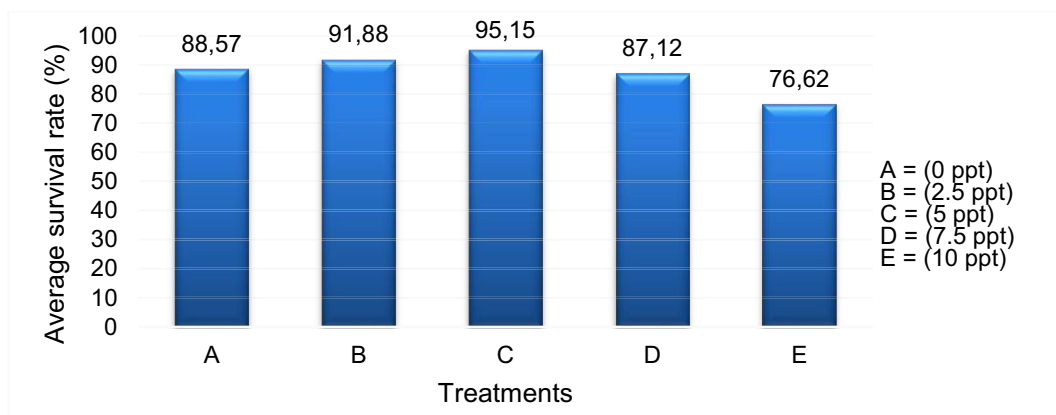


Figure 4. Percentage of survival rate of larvae

Figure 4 shows that the higher the salinity, the higher the survival rate of tilapia larvae, but at a certain point the survival rate reaches the maximum value and then decreases. Treatment E (salinity 10 ppt) resulted in an average survival rate of 76.62%. This result is the lowest result among treatments, but is still high. Treatment E produces the lowest survival rate because the higher the salinity, the greater the energy needed for the osmoregulation process, so that the energy that should be needed for growth and maintaining its survival is reduced. This is in accordance with the remark of Rahim *et al.* [16] argued that the higher the salinity, the higher the mortality of tilapia fish seeds. Because if high salinity is not accompanied by the ability to osmoregulate, it will result in the death of tilapia. Tilapia survival is influenced by its osmoregulation ability. Although tilapia have a natural habitat in freshwater, tilapia can adapt to high salt levels.

The treatment that resulted the highest survival rate was treatment C (5 ppt salinity) with an average of 95.15%. Treatment C resulted in the highest survival rate because at this salinity the fish was in an *isoosmotic* state. At a salinity of 0 ppt - 10 ppt, the survival rate is still relatively high. This is because tilapia is classified as *euryhaline* fish. According to Hena *et al.* [17] informed that tilapia is a *euryhaline* fish that can live in a wide salinity range from freshwater to sea water, so it has a higher tolerance for salinity than other fish.

Cells that play a role in the osmoregulation process of tilapia are chloride cells. According to Sholikhah and Widyaningrum [18], chloride cells are cells that regulate the exchange of salt and water in the body with their environment, and play a role in the removal of nitrogenous wastes. According to Arrokhman *et al.* [19], if the fish are in a higher salinity environment, the number of chloride cells will increase and vice versa if they are in an environment with a lower salinity. These short-term changes can lead to long-term changes in gene expression.

The data of variance above shows that the results obtained are F count > F5% and F1%. Based on these results, it can be concluded that the survival of tilapia in different salinity media containing the enzyme bromelain 0.1223 μmol has a very significant effect. The regression graph of the effect of salinity media on tilapia survival is presented in Figure 5.

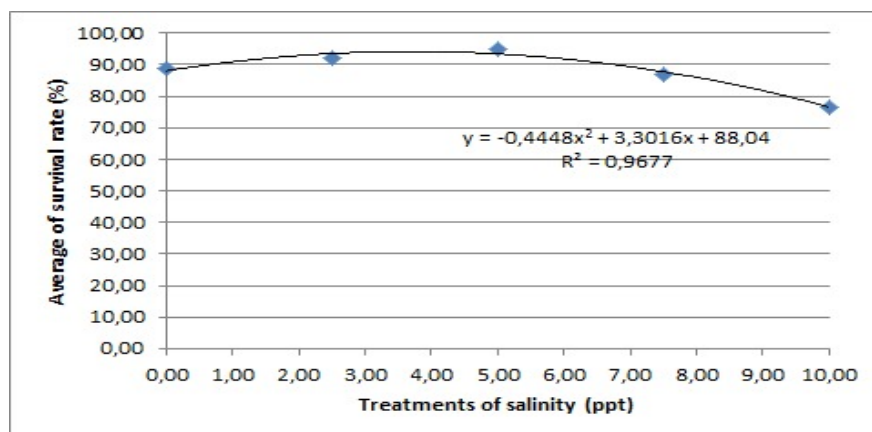


Figure 5. Regression graph of the effect of salinity media on tilapia survival rate

The graph results obtained in the form of a quadratic curve so that the higher the salinity, the higher the hatching rate obtained. However, at a certain point the hatching rate reaches a peak and the yield will decrease further.

3.4 Water quality

Water quality conditions are one of the important factors that support the success of the hatching process. Good environmental conditions will produce high hatching rate, on the other hand, if the conditions are not good, the hatching rate will also decrease. Water quality measured in this study included temperature, pH and low oxygen. The range of water quality at the time of the study can be seen in Table 1.

Table 1. Range of water quality

Parameters	Range	Literature
Temperature (° C)	24 – 27	25 – 29 [20]
pH	7.2 – 8	6.5 – 8 [21]
DO (ppm)	5 – 6,5	≥5 [6]

4. Conclusions

Based on the research results, the conclusions are the difference in salinity in media containing the enzyme bromelain of 0.1223 μmol has a very different effect on hatching rate of tilapia eggs. The results of the treatment that produced the best hatching rate were treatment C with a salinity of 5 ppt. The highest average hatching rate value was 88.89% + 3.85 with the equation $y = 64.127 + 7.4095x - 0.6476x^2$ and $R^2 = 0.794$. From this equation, the maximum X (salinity) value is 5.71 ppt.

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